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RESEARCH

Energy Communities' Archetypes and Operational Scenarios Using Agents and Embedding Heating and Transportation

Adela Bâra[1] · Simona-Vasilica Oprea[1] · Corina Murafa[2]

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Abstract

Energy Communities (ECs) in ex-communist countries face cultural and technical challenges that have historically hindered their adoption. This paper examines various urban and rural archetypes and operational scenarios for ECs in these regions. We propose a methodology for selecting the best scenario based on factors such as self-sufficiency, investment payback period, cost savings, fair value share allocation, and CO_2 emissions. EC's members are modeled as agents using *Mesa* and *PyMarket* Python libraries. Our analysis indicates that integrating heating and transportation within ECs significantly enhances efficiency, leading to substantial benefits for residents. Cost savings range from 20% when replacing gas heating with heat pumps to 59% with electric transportation, increasing EC attractiveness. Thus, we demonstrate that these integrated scenarios are feasible and viable, with payback periods between 1 and 9 years for medium solar photovoltaic (PV) systems (up to 200 kWp). Furthermore, environmental benefits are notable, with CO_2 emissions reduced by 25% when replacing gas heating and 60% with transportation. This paper provides evidence of the practicality of these scenarios, highlighting their potential to improve energy management and community satisfaction.

Keywords Energy community · Ex-communist countries · Archetypes · Operational scenarios · Renewables · Heating · Transport · Sector coupling

1 Introduction

Energy communities (ECs) are emerging in both urban and rural contexts, adapted to local resources and infrastructural challenges. In rural areas, they enhance energy autonomy through renewable sources like biomass, wind, and solar, supported by strong social cohesion and cooperative models. Urban ECs focus on solar PV generation, but increasingly also on retrofitting inefficient residential blocks, particularly those from the communist era, with energy-efficient technologies and rooftop usage, enabling collective energy generation and governance to reduce dependence on external suppliers. The problem which triggered the exploration of urban and rural archetypes of ECs in Romania, coupled with different operational scenarios of increasing complexity, is the lack of clearly established ECs in Central and Eastern Europe (CEE) at large and in Romania in particular.[1] While ECs have spread significantly in Western Europe, with 4% of Europe's citizens being a member in one of the 9000 + existing communities [23], the movement has been slow to pick up in Eastern Europe, despite increasing awareness over this new market model and despite the transposition of relevant European legislation. While researchers anticipate an exponential growth of energy communities in Europe by 2050,[2] there is limited know-how on the "value-for-money" of energy communities [38] and some authors purport that the

✉ Simona-Vasilica Oprea
simona.oprea@csie.ase.ro

Adela Bâra
bara.adela@ie.ase.ro

Corina Murafa
corina.murafa@fabiz.ase.ro

1 Department of Economic Informatics and Cybernetics, Bucharest University of Economic Studies, No. 6 Piaţa Romană, Bucharest, Romania

2 Faculty of Business Administration, Bucharest University of Economic Studies, No. 6 Piaţa Romană, Bucharest, Romania

1 Energy Communities: an Overview of Energy and Social Innovation—DocsLib.

2 Report: The potential of energy citizens in the EU—Greenpeace European Unit.

cost-effectiveness of the model can be assured only through subsidization [7]. The complexity of the organizational setup, ambit, and scale of ECs across the world makes it all the more difficult for early-stage geographies, such as CEE and/or Romania, to understand the feasibility (economic, environmental, technological) of different types of ECs and thus the decision-making process in setting up an EC becomes increasingly cumbersome. Thus, through the analysis contained in this article, the academic and policymaking communities are provided with clear figures, primarily on the economic feasibility (upfront investment costs, cost savings, etc.) of increasingly complex archetypes and scenarios of ECs—from very low-capacity ones covering strictly common electricity consumptions to ECs in which all consumptions of members (heating, electricity, transportation, etc.) are covered.

The motivation to undertake this research effort was also due to the lack of a clear modeling framework for assessing the feasibility of different EC models, using real grassroots data collected from market statistics and actual consumption data of potential ECs. While existing literature on the matter of ECs has become increasingly complex and dense, particularly in the last couple of years, most of the current literature on the matter of ECs uses neither survey data nor case studies to determine the advantages of ECs or the obstacles (legislative, technical, social, etc.) to creating and implementing ECs. A smaller stream of existing literature surveyed different types of business models that ECs are developing and innovating towards [19, 38], but does not formulate complex scenarios of increasingly complex coverage of different energy needs (including transportation and heating) by the energy communities proposed. In general, literature investigates multipurpose energy communities, covering exclusively the electricity needs of users [18, 59], although case evidence shows greater emissions reduction potential by other types of energy communities, such as hot water energy communities [50] or thermal energy communities [36]. The (under)development of ECs in ex-communist countries remains an understudied topic in the literature [16]. Existing case studies do not model their feasibility in the region, but rather pinpoint factors hampering their development, such as the lack of energy market deregulation and domination of state-owned enterprises in the market [65], excessive bureaucracy [29] or use a very limited definition of an energy community, excluding the citizen participation dimension [66].

The context of ex-communist countries in modeling of ECs is central to our research. This context matters for several reasons, such as (a) institutional and cultural legacy. Ex-communist countries, including Romania, have a legacy of top-down energy systems and enforced collectivism, which has resulted in low levels of civic engagement and limited trust in cooperative models. This sociocultural background directly influences agent behavior in our modeling, particularly in terms of participation rates, trust, and willingness to invest in collective infrastructure; (b) infrastructure constraints. Many urban areas still operate with aging, inefficient residential infrastructure from the communist period. This affects the technical modeling of EC archetypes, especially in retrofitting scenarios where energy inefficiencies and limited rooftop space constrain photovoltaic deployment; (c) regulatory and market readiness. These countries generally lag behind Western Europe in energy market liberalization, smart metering infrastructure, and incentives for prosumers. This shapes the operational scenarios modeled, particularly with regard to interaction with local flexibility markets (Scenario F) and assumptions about energy pricing and returns; (d) financial feasibility and risk perception. Due to lower average incomes and higher perceived risks, the economic feasibility modeling must include more conservative investment thresholds and longer payback period tolerance compared to Western EU contexts. These assumptions influence scenario selection and prioritization. Therefore, the context of ex-communist countries is an essential variable that informs the design, simulation, and feasibility assessment of ECs across multiple dimensions in our research.

Thus, the aim of our paper is to support the development of ECs in the region by modeling rural and urban archetypes of ECs and developing operational scenarios of cascading complexity, using datasets as detailed in Section 4 collected from prospective community settings and synthetically generated, trying to gauge the economic feasibility of the model without subsidization and with citizen participation. The structure of our paper begins with an examination of relevant literature to clearly describe the main findings of major current research streams on ECs and to emphasize the research gap the paper is intending to fill. Existing literature is non-ambiguous on the financial and economic benefits of ECs, examining case studies across a wide geography, but there is a highly critical stream of literature on the social impact claims that policymakers hold with regards to ECs, asserting they boost citizen engagement, energy democracy, and social cohesion. As a result, more empirical analyses are obviously needed, yet they are hard to do in regions such as CEE and in countries such as Romania, where the early-stage development of this new market model does not allow for ex-post analyses. The following section outlines the research methodology, distinguishing between urban and rural settings through three EC archetypes (two urban: one in a new and one in an old residential area and one rural). Six progressively complex scenarios are developed, ranging from coverage of common area consumption to full energy autonomy and market integration. Key enabling factors for each scenario are identified. Each scenario is assessed in terms of initial community payments, revenues from surplus electricity injected into the grid, and residual payments.

Capital costs are estimated, and formulas are introduced for calculating Value Share (VS), reflecting internal surplus distribution benefits. A unified approach for converting diverse energy forms (heating, transport) into electricity equivalents is also presented. A strategy for selecting the optimal EC configuration is proposed, based on cost savings, grid dependence, self-sufficiency, fairness, and CO_2 reduction. Section 4 details simulations of the scenarios using 14 performance indicators, incorporating seasonal sensitivity. These simulations span 1 year and involve 114 apartments. The study concludes that integrating electric mobility significantly improves economic viability, whereas scenarios involving heating demand higher upfront investments and longer payback periods.

The evidence brought forward in the paper demonstrates the strong feasibility of the model in CEE and further that the more complex the services offered by the community, despite higher upfront development costs, the higher the economic and environmental benefits are, and thus the more sustainable the model. The paper will hopefully serve to enrich knowledge about ECs in CEE and to contribute to a dynamic policy landscape in the region, where policymakers, citizens, and local authorities are trying to co-design supporting legislative frameworks for the emergence of this new form of energy market participation.

2 Literature Review

The advantages and obstacles of ECs are examined in distinct literature streams. When it comes to the advantages of ECs, one stream focuses on the economic and technical aspects, while the other focuses on the social and community advantages. Authors who focus on the former stress that local ECs contribute to local economic development, creating jobs and raising tax revenue [43, 56], energy bill savings of 62% and energy provision cost reduction up to 26% [26] and a ROI of 2 to 7 years at self-consumption levels ranging between 39 and 65% [57].

Many studies model the economic advantages of ECs vs. individual prosumers, in the sense of reduced energy costs, but also increased investment value. By using a mixed-integer linear optimization model developed to maximize the net present value over a time horizon of 20 years, [27] demonstrate that the actual added value of an EC, especially when some of its members have large rooftops available for solar PV installation, is higher than the value generated by individual consumers. [48] prove not only the reduction of energy costs and grid stability by analyzing pilot ECs across Europe, but they also focus on environmental and social advantages, as perceived by all stakeholders inside and outside the community. Another economic advantage demonstrated in academic literature investigates the increased mobilization of investments under an EC model [11, 17], including by boosting individual investment decisions towards storage/battery options [21].

Others reveal the fact that ECs and the underlying system of peer-to-peer (P2P) trading minimize transmission losses and thus increase the stability of supply [2, 78]. Moreover, [40] emphasize that an EC model is particularly beneficial for businesses, while [77] and [76] believe that microgrid-blockchain innovation is particularly suitable to boost the economic advantages of ECs. The techno-economical advantage of reducing grid imbalances is emphasized as well in different sources [53, 54, 58]. The economic advantages spread beyond the actual EC to the wider region in which the EC initiative is located [46, 69].

Authors who focus on the latter, namely non-economic or technical benefits, point out benefits such as innovation diffusion [15], [61], including by demonstrating that EC members, once accustomed to the practice of citizen participation in the energy markets, rolled out new, community-based business models, such as community car sharing for EVs [63]. [9] point out that EC members "confirm the importance of member involvement (process), shared interests (interest) and a sense of togetherness (social interaction)" while other authors are more skeptical with regard to the potential of ECs to drive democratic citizen engagement [37]. Energy independence increase is seen as another benefit of ECs [1, 24] and thus P2P trading in an EC model becomes very attractive to consumers [33], especially for environmentally responsible and technologically savvy consumers [32], thus making the whole business of EC very attractive as a business model, especially with the gradual phaseout of feed-in tariffs and other subsidy schemes in many Member States [47].

Another clear contribution ECs seem to be making is towards increasing the share of renewables in the energy mix. ECs are seen as tools for increasing social cohesion and community empowerment [60, 67, 68]. When it comes to the matter of social cohesion, nonetheless, studies point out that this claim is not backed by evidence and that it is in fact not confirmed by data either [20, 35], while other streams of research point out that other types of impact are not substantiated by data either [10]. Systematic literature reviews of the evidence behind the social value creation of ECs done by [12] conclude that "positive social impact of ECs can only be assumed" and that "the underlying narrative that ECs have an intrinsic social value needs to be reevaluated" Furthermore, in countries where decentralized energy initiatives have been championed for a long time, i.e. Germany, ECs now find themselves in a situation to reinvent themselves and "provide solutions to the systemic challenges of the energy transition process" [8]. Authors even call for a reinvention and reclassification of their taxonomy [52]. Despite this criticism, clear evidence exists that ECs increase citizens' preoccupation with energy efficiency and reduce consumptions [39, 42, 71].

Literature focused on the obstacles of ECs shows that the main obstacles towards the deployment of ECs at scale in Europe are vague legislative definitions, complex regulation, and the lack of adequate incentive schemes [14, 75], such as feed-in-tariffs [76], but especially frequent policy changes [51]. Case studies from Portugal, Spain, Norway, and Poland show that, indeed, legislative obstacles bring about the greatest pain points. Other obstacles are the lack of targets for the development of ECs in national policies, the resilience of communities in adopting certain renewable technologies, skepticism about cooperative and collective structures, as well as the lack of knowledge about the benefits of ECs on national energy systems [6, 34]. Literature points to the process of transposition of RED II as a great catalyst for the formation of ECs [44]. Interviews undertaken with members of ECs, interpreted by using analytical hierarchical processes, demonstrate that the "regulatory complexity and legal limitations emerge as the top-ranked barriers" [22], while studies dealing with a similar category of respondents, by using different methods, namely integrated multi-criteria decision-making approach to develop decision maps, conclude that lack of community awareness with regards to technology is an important grassroots barrier for ECs [73]. More recent studies point out that, while there is growing interest in ECs from both municipalities (especially municipal companies) and private actors, "citizen participation is lacking across cases" [13]. All in all, the main conclusion of academic literature on barriers for ECs, in diverse geographies, is that such barriers are diverse, but that technological maturity, political will, and incentives are the dominant barriers [3].

Literature dealing with the business models employed by ECs in different geographies and across a variety of technologies shows that there is a financial viability of the EC business model, i.e., cost reduction of up to 26% against business-as-usual cases [26] or profit gains under worst case scenarios, in the case of a woody biomass community [41] or in the case of a thermal EC-based multi-dimensional business model framework [30]. However, there are also studies which demonstrate, using large-N samples, that the economic viability of EC business models is not yet proven [64], with several factors, such as the number of members, being critical factors in determining the financial sustainability of ECs [38]. In all cases, what characterizes ECs from a business model perspective is innovation/the development of new business models (e.g.: ECs becoming distribution system operators or engaging in multiple activities of community-to-community trading, according to [28]. Authors [74] warn that different EC business models are needed for different policy objectives (e.g., increasing renewables capacity, mobilizing private capital, empowering consumers, etc.). For instance, bottom-up and P2P market business models empower consumers mostly [49], while large-scale prosumers included in the EC model significantly enhance the grid stabilization value proposition of ECs [45]. Empirically though, the diversity of business models in the EC is limited, traditional self-consumption place-based communities being dominant, "while business models involving differentiated services as demand flexibility, aggregation, energy efficiency and electric mobility are still scarce" [25].

Given the knowledge landscape on ECs, as detailed above in the literature review section of our paper, two gaps in existing literature are being filled. Firstly, the lack of analytical studies focused on ECs is addressed, with data and insights collected from the region, at the grassroots level. Second, the lack of integrated models to test the feasibility of different scenarios of increasingly complex multipurpose ECs from a diversity of advantages (economic, environmental, technological—i.e. grid dependence) is addressed. The framework developed can serve as a fertile testing ground for looking into a wide range of archetypes of ECs, depending on their interests, needs, and available resources.

3 Methodology

ECs are being developed in both urban and rural settings, each tailored to harness local strengths and address specific energy inefficiencies inherited from centralized, often outdated systems. In rural areas, ECs focus on leveraging local natural resources and strong social bonds to enhance energy independence and sustainability. They might utilize biomass from agricultural waste, alongside wind and solar resources, to create a robust local energy conversion ecosystem. The communal nature of rural life also facilitates cooperative approaches to energy management, such as shared solar installations or community-managed microgrids, which can significantly improve resilience and reduce dependence on unreliable national grids.

Urban environments, particularly in large cities with existing residential blocks from the communist era, present different challenges and solutions. These high-density blocks are typically marked by significant energy inefficiencies. Retrofitting them with modern energy-efficient technologies, such as improved insulation, energy-efficient windows, and advanced heating systems, becomes a priority. Additionally, the flat rooftops common in these blocks are ideal for installing solar panels, potentially enabling buildings to become partially or fully energy self-sufficient. Moreover, residents in these urban blocks may form cooperatives to invest collectively in energy improvements, manage their generation capacities, and negotiate better terms with energy suppliers, thus leveraging group buying power and shared investments for greater energy autonomy.

Three EC archetypes may coexist: two urban ECs—(1) existing blocks in the urban areas, large cities, and (2) new residential districts—and one rural EC. Several evolving scenarios can be imagined. The EC developing scenarios investigated in this paper are presented in Fig. 1.

New residential districts represent a forward-looking approach to urban planning in ex-communist countries, characterized by suburbanization [72], re-urbanization, and regeneration [31]. These districts are often designed from the ground up with modern standards of energy efficiency and sustainability. They incorporate smart energy systems, including advanced metering and dynamic energy management systems, to optimize usage and generation. A mix of integrated RES, such as solar, wind, and geothermal, along with cutting-edge technologies like heat pumps (HP), EV charging stations, and energy storage systems, is planned. The design of these districts also prioritizes sustainable urban development, maximizing natural lighting and heating, and integrating green spaces to enhance the living environment.

The exploration of evolving scenarios within ECs presents a detailed roadmap towards increasing levels of self-sufficiency and integration with local energy markets. Each scenario builds upon the previous one, gradually incorporating more aspects of electricity and thermal energy consumption and production within the community (as in Table 1).

These scenarios represent a progressive build-up towards an integrated, sustainable, and economically beneficial energy ecosystem within residential communities in ex-communist countries. Each step increases the EC's energy independence and also enhances its resilience and sustainability, making a significant contribution to broader environmental and economic goals.

The transition towards ECs in ex-communist countries involves a profound shift in technological adoption and also in overcoming cultural and psychological barriers associated with historical experiences of enforced collectivism and with persistently low levels of civic engagement [62] and low willingness to engage in collective action [70]. This legacy of skepticism makes the implementation of ECs particularly challenging [55]. However, by extending the scope of ECs to include critical services like heating and transportation and transitioning these to RES, the tangible benefits become more apparent, potentially persuading more community members to participate.

One of the most compelling advantages of expanding ECs to include heating and transportation is the potential for significant cost savings. Transitioning from traditional fossil fuels to RES-based systems for heating and EVs for transportation reduces dependency on expensive and volatile energy sources (gas, wood, gasoline). This offers direct financial relief to individuals and also enhances the overall energy security of the region. The shift to local energy generation through ECs reduces the EC's vulnerability

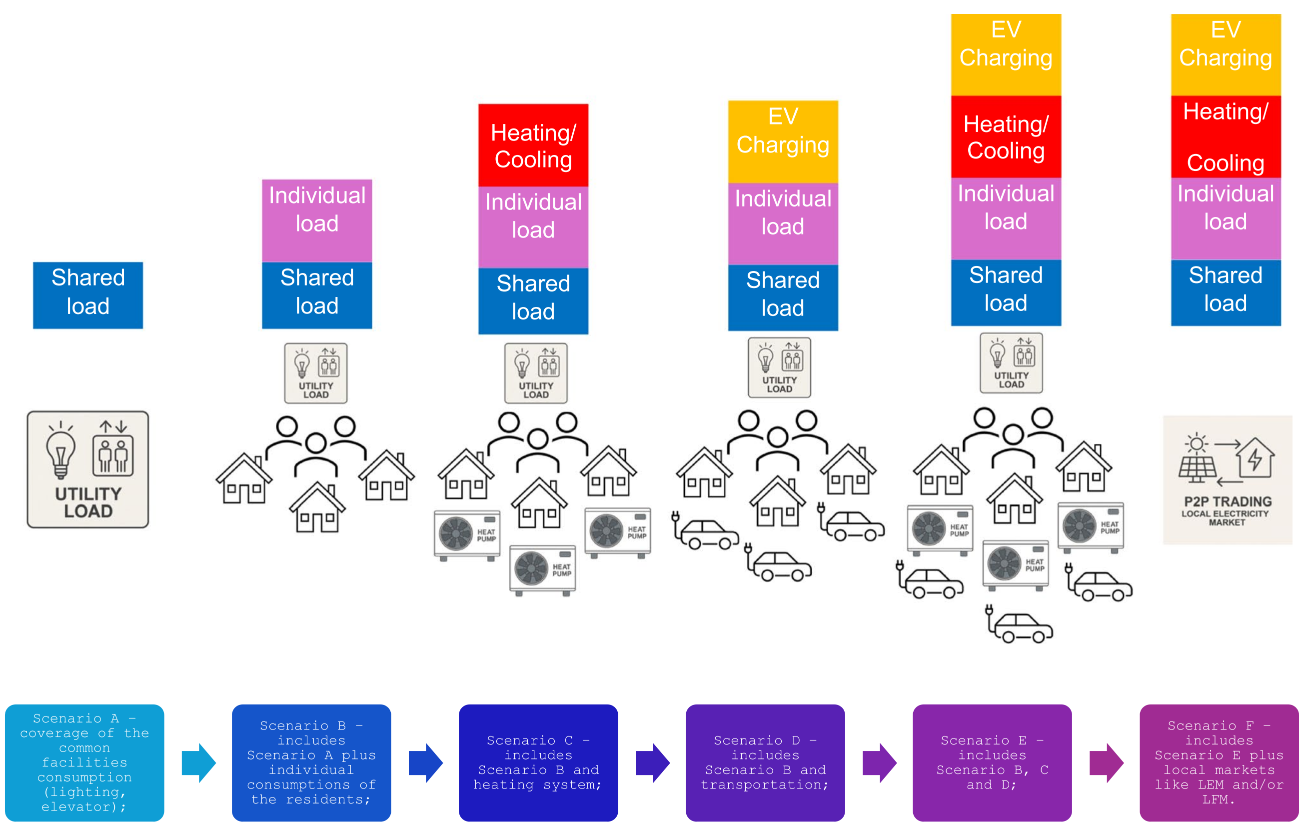

Fig. 1 ECs archetypes and evolving scenarios

Table 1 Description of the developing EC scenarios

Scenario	Depiction
Scenario A	Coverage of common facilities consumption. In this initial scenario, the focus is on covering the energy needs of common facilities within a residential community. This includes lighting in shared areas such as hallways, lobbies and outdoor lighting, as well as the power needed for elevators. The objective here is to manage and reduce the communal energy costs, potentially through solar panels installed on building rooftops or other communal energy generating systems. This scenario sets the foundation for community energy independence at a basic level
Scenario B	Includes Scenario A plus individual consumptions of the residents. Building on Scenario A, Scenario B extends the energy coverage to include individual consumption within each residential unit. This encompasses all domestic energy uses such as lighting, appliances and electrical devices in individual apartments or homes. This might involve more extensive solar installations or small-scale wind energy solutions, plus individual metering and possibly energy storage systems to ensure that generated energy covers peak demand times and reduces grid dependence
Scenario C	Includes Scenario B and heating system. Scenario C adds an important component to the energy management system: the heating of residential units. Heating typically represents a significant portion of energy consumption, especially in colder climates common in many ex-communist countries. Integrating heating into the community's energy plan might involve the installation of centralized geothermal heat pumps, large-scale biomass boilers or the incorporation of district heating systems that use waste heat from local industries
Scenario D	Includes Scenario B and transportation. This scenario integrates electric transportation solutions into the energy planning of the community. This might involve providing energy for charging EVs and other devices based on batteries, using excess energy generated from the community's RES. Infrastructure such as EV charging stations would be included in communal areas, encouraging residents to transition to EVs and thus reducing the community's overall carbon footprint
Scenario E	Includes Scenarios B, C and D. Scenario E represents a comprehensive integration of all previously mentioned aspects: individual and common facility energy needs, heating and transportation, into a unified community energy system. This scenario maximizes the self-sufficiency of the community, making it nearly independent from the external grid. The complexity of managing such a diverse array of energy sources and needs would likely require sophisticated energy management systems with smart grid capabilities
Scenario F	The most advanced scenario, Scenario F, extends beyond self-sufficiency to interaction with local energy markets such as Local Energy Markets (LEM) and Local Flexibility Markets (LFM). This scenario envisages the community not only covering its own energy needs but also actively participating in the broader energy market. This involves selling excess energy to the grid or other users and purchasing energy when it is more cost-effective. Participation in LFM would allow the community to offer services like Demand Response (DR), where the community can adjust its energy usage patterns in real time in response to grid/power system needs

to external disruptions and price fluctuations, presenting a strong case for energy independence that may resonate deeply in regions with unstable energy supplies.

Building trust within the community is equally important. Educational programs that explain both the practical and social benefits of ECs may help mitigate distrust. Additionally, educational campaigns funded and largely disseminated by the government can help dispel longstanding misconceptions about cooperative ventures.

3.1 Assessment of the Proposed Scenarios

3.1.1 Scenario A

The initial payment of the community ($Pay_{EC,0}$) includes the shared electricity cost for common assets (e.g., lighting, elevator, utilities, denoted by C^t_{EC}), the sum of the individual electricity costs (C^t_i), the annual cost for heating ($CostH_i$) using gas, wood or other conventional sources, and the annual transportation cost ($CostT_i$) based on fossil fuel use. Electricity consumption and generation are modeled with high temporal granularity and calculated using Time-of-Use (ToU) tariff rates for each time interval t. These intervals may correspond to various resolutions (e.g., hourly, 15 min or 30 min), depending on the metering infrastructure of the community. The total electricity cost is aggregated over the full annual profile using these sub-hourly measurements. In contrast, conventional heating and transportation costs are considered as total annual values, based on estimated yearly consumption and average fuel prices:

$$Pay_{EC,0} = \sum_t \left(C^t_{EC} + \sum_{i \in EC} C^t_i \right) \times ToU^t + \sum_i \left(CostH_i + CostT_i \right). \quad (1)$$

The payment after A is calculated as the difference between the cost of the demand covered from grid at ToU and the revenue from the surplus injected into the grid using Feed-in Tariff (FiT) as in Eq. (2). Additional costs may be added for maintenance of the PV panels:

$$Pay_{EC} = \sum_t \left(D^t_{EC} \times ToU^t - S^t_{EC} \times FiT^t + \sum_{i \in EC} C^t_i \times ToU^t \right) + \sum_i \left(CostH_i + CostT_i \right), \quad (2)$$

where D_{EC}^t is the demand covered from the grid and the S_{EC}^t is the surplus injected into the grid in case the shared generation (G_{EC}^t) exceeds the shared consumption:

$$D_{EC}^t = \begin{cases} C_{EC}^t - G_{EC}^t, if C_{EC}^t > G_{EC}^t \\ 0, otherwise \end{cases}, \tag{3}$$

$$S_{EC}^t = \begin{cases} G_{EC}^t - C_{EC}^t, if G_{EC}^t > C_{EC}^t \\ 0, otherwise \end{cases}. \tag{4}$$

Let's denote by $Cost_{EC}$ the investment cost for transition towards Scenario A. It is composed of the investment cost for procurement and installing the PV system: PV panels, inverter, additional equipment, labor and permits and it is calculated based on the rated power of the systems (P_{PV}) multiplied by the cost per unit (UC_{PV}):

$$Cost_{EC} = P_{PV} \times UC_{PV}. \tag{5}$$

The power installed in the PV system (P_{PV}) can be determined by dividing the total annual electricity demand to be covered ($\widehat{C}_{EC}^{365}$) by the specific PV yield (Y_{spec}) expressed in kWh/kWp/year for the location as indicated by the Global Solar Atlas.[3] The value already includes system losses (inverter efficiency, temperature effects, soiling, etc.) and thus offers a realistic estimate for required PV sizing:

$$P_{PV} = \frac{\widehat{C}_{EC}^{365}}{Y_{spec}}. \tag{6}$$

P_{PV} may be limited by the size of the PV system that can be installed on the available surface (roof or land), but given that photovoltaic (PV) panels now achieve a rated power exceeding 550 kWp per panel, it is possible to install higher rated power systems on existing surfaces.

The payback period of the investment cost (PBP_{EC}) depends on the installed power and the difference between the initial and the final payment of the corresponding scenario:

$$PBP_{EC} = \frac{Cost_{EC}}{Pay_{EC,0} - Pay_{EC}}. \tag{7}$$

3.1.2 Scenarios B and C

In these scenarios, the generation of the PV systems is used to cover the shared consumption of the community and the individual consumption of the members, including heating for Scenario C. Let's denote by C_{sh}^t the shared consumption and with G_{sh}^t the shared generation allocated to each member supposing that an equal distribution is used:

$$C_{sh}^t = \frac{C_{EC}^t}{m}, \tag{8}$$

$$G_{sh}^t = \frac{G_{EC}^t}{m}. \tag{9}$$

In case the members' contribution is not equal, a proportional quota can be used to calculate the corresponding values of C_{sh}^t and G_{sh}^t for each member. The individual demand and surplus are calculated using Eqs. (10–11) considering the own consumption/generation plus the shared consumption/generation:

$$D_i^t = \begin{cases} C_i^t + C_{sh}^t - \left(G_i^t + G_{sh}^t\right), if C_i^t + C_{sh}^t > G_i^t + G_{sh}^t \\ 0, otherwise \end{cases}, \tag{10}$$

$$S_i^t = \begin{cases} G_i^t + G_{sh}^t - \left(C_i^t + C_{sh}^t\right), if G_i^t + G_{sh}^t > C_i^t + C_{sh}^t \\ 0, otherwise \end{cases}, \tag{11}$$

where G_i^t represent the individual generation of member i in case he/she installed own PV systems and became a prosumer in the community and C_i^t represents the individual consumption composed by: the individual load of the member expressed by CP_i^t (lights, refrigerators, surveillance cameras, multimedia appliances, washing machines, boilers, dryers), and heating/cooling consumption expressed by CH_i^t in case of scenario C:

$$C_i^t = CP_i^t + CH_i^t. \tag{12}$$

The individual payment of the member i is calculated as in Eq. (13) as a difference between the cost of the demand and the revenue of the surplus:

$$Pay_i = \sum_t \left(D_i^t \times ToU^t - S_i^t \times FiT^t\right). \tag{13}$$

The demand of the community is the sum of the individual demand partially covered by the individual surplus, while the surplus of the community is the sum of the individual surplus partly consumed by the individual demand:

$$D_{EC}^t = \begin{cases} \sum_{i\in m} D_i^t - \sum_{i\in m} S_i^t, if \sum_{i\in m} D_i^t > \sum_{i\in m} S_i^t \\ 0, otherwise \end{cases}, \tag{14}$$

$$S_{EC}^t = \begin{cases} \sum_{i\in m} S_i^t - \sum_{i\in m} D_i^t, if \sum_{i\in m} S_i^t > \sum_{i\in m} D_i^t \\ 0, otherwise \end{cases}. \tag{15}$$

Thus, the payment of the community is calculated as the difference between the cost of the demand covered from grid at ToU and the revenue from the surplus injected into the grid at FiT:

$$Pay_{EC} = \sum_t \left(D_{EC}^t \times ToU^t - S_{EC}^t \times FiT^t\right). \tag{16}$$

[3] https://globalsolaratlas.info

Since the individual demand of the members is partially covered by other members who are in surplus, the total payment of the community is lower than the sum of the individual payments. Therefore, the difference, known as Value Share (VS) is obtained as a gain of joining the community and sharing the surplus internally. VS is obtained as follows:

$$VS_{EC} = \sum\nolimits_{i\in m} Pay_i - Pay_{EC}. \tag{17}$$

The VS can be allocated to the members using different methods such as Equal Quota, Marginal Contribution, or Consumption/Generation-based models, as detailed and analyzed in [4]. After VS allocation, the final payment of the members is obtained as follows:

$$Pay_i \leftarrow Pay_i - VS_i. \tag{18}$$

Related to the investment cost for scenarios B and C, the installed power of the PV system should consider a higher consumption for the entire year ($\widehat{C}_{EC}^{365}$) for covering the increased demand for electricity. In case of scenario C, the demand for electricity used for heating during the cold period can be estimated by converting the consumption of conventional sources (gas or wood) to electricity. Let's denote by QH_i the heat energy requirement per hour for a household. It is calculated as follows:

$$QH_i = V_i \times \Delta T_i \times k, \tag{19}$$

where V_i represents the volume of the house (surface multiplied by height), ΔT_i is the temperature difference between inside and outside, and k is the insulation coefficient of the house. The insulation coefficients used in Eq. (19) were based on standard values reported in the European building energy efficiency literature.[4] Specifically, well-insulated, $k \approx 0.2$ W/m^3/°C (new buildings or retrofitted with insulation); medium-insulated, $k \approx 0.5$ W/m^3/°C; and poorly insulated, $k \approx 1.0$ W/m^3/°C (typical of old, unrenovated residential blocks).

To obtain the electricity consumption necessary to cover the heating requirement, an efficiency coefficient is applied to transform QH_i into CH_i. In the case of electric heating, the coefficient is equal to 1, for gas heating it varies between 0.8 and 0.95, while in case of heat pumps the Coefficient of Performance (COP) is used. The typical values of COP for heat pumps usually range between 3 and 4 but can be higher under ideal conditions. In the simulations, we assumed a constant average COP of 3.2, based on typical performance for modern air-to-water heat pumps in moderate continental climates.[5] This average value reflects seasonal performance rather than a detailed COP profile, due to the annual modeling approach and limited hourly COP data availability.

Two options are available for scenario C: centralized heating with a heat pump that serves the entire community, and each member receives a quota based on his/her consumption or individual heating when each member installs heating pumps or use electric heaters (convectors, radiators or thermal plasmas) for self-heating. Both options can be reflected by CH_i^t, but the investment cost in case of individual heating is supported by the consumer and not shared between the members.

The investment cost for scenario C using centralized heating includes the investment cost for replacing the conventional sources (procurement and installing the heating pumps and additional equipment). In the simulation section, an average cost is considered, and the period for recovering the total investment is calculated accordingly, using Eq. (7).

In case the conventional sources for heating are not entirely replaced, the payment of the community includes the corresponding cost for these sources ($CostH_i$), but with a lower value.

3.1.3 Scenarios D and E

In these scenarios, the EVs are considered to replace partially or entirely the conventional transportation based on petrol and diesel. Therefore, the values of the $CostT_i$ will be lower or even 0 for some members. Instead, the initial investment cost is composed of the investment cost of the PV system with a higher capacity than Scenario A or B. Two options are available for EV charging stations: centralized Level 3 (DC fast charging) stations serving the community, or individual charging using Level 1 or Level 2 equipment. Level 1 chargers typically draw 1.2 kW from a 120 V outlet, and Level 2 chargers provide around 7.2 kW from a 240 V outlet. The investment costs for these stations are supported by each member and are not considered as community costs. In contrast, Level 3 (DC fast) chargers can deliver between 50 and 350 kW, depending on the system design. For a community (block of apartments or houses), installing a Level 3 charging station is more complex than installing Level 1 or Level 2 chargers. The investment cost for Level 3 charging stations is shared between the community members and includes the procurement cost of the equipment, installation and permit costs. In our simulation, we assume a shared Level 3 station with a rated charging power of 80 kW for community-scale deployment. The energy demand for charging is allocated to individual members via digital identification (e.g., cards or PINs) and added to each member's electricity load profile. Therefore, the consumption for charging the EV in both options can be expressed by CEV_i^t that is added to the total consumption of member i in Eq. (12).

4 https://www.iso.org/standard/65696.html

5 https://www.ehpa.org/news-and-resources/publications/ehpas-annual-report-2023/

To assess the electricity consumption necessary to cover transport needs, the annual distance to be covered (km^{365}) representing the number of kilometers per year is considered and an average EV consumption (EC_{EV}=15kWh/100 km), adjusted by the charging efficiency $\eta_{EV} = 0.90$:

$$\hat{C}_{EV}^{365} = \frac{EC_{EV}}{\eta_{EV}} \times \frac{km^{365}}{100}. \tag{20}$$

This estimation is added to $\hat{C}_{EC}^{365}$ in Eq. (7) to determine the PV rated power required to satisfy the total consumption need.

Some additional costs in the case of scenarios D and E may include the annual maintenance costs for the EV charging stations. The total payment of the community, the individual payments and value share are calculated using Eq. (13–18), considering that the consumption C_i^t in scenario D has $CH_i^t = 0$.

3.1.4 Scenario F

This scenario includes LEM that enables participants to trade their surplus and deficit on the community at more convenient prices compared to FiT in case of surplus and ToU in case of deficit. Let's denote by QB_i^t and QS_i^t the quantities for deficit and surplus traded on LEM and with P^t the price on LEM, supposing that a Uniform Price mechanism is applied for market clearing. Thus, the individual payments are lower compared to previous scenarios and the Pay_i from Eq. (13) becomes:

$$Pay_i = \sum_t \left[\left(QB_i^t \times P^t + \left(D_i^t - QB_i^t \right) \times ToU^t \right) - \left(QS_i^t \times P^t + \left(S_i^t - QS_i^t \right) \times FiT^t \right) \right]. \tag{21}$$

The total payment of the community is calculated using Eq. (16) and the value share is obtained as in Eq. (17). The VS in this scenario is lower than the previous scenarios since the sum of the individual payments is lower, while the Pay_{EC} remains unchanged.

3.2 Setting the Strategy for Selecting the Best Scenario That Fits the Community Needs

The strategy for selecting a specific scenario considers first the difference between the final and initial payments and the period for recovering the investment costs. But not only monetary aspects should be included when deciding to adopt a specific scenario. Aspects related to fairness, grid dependence, self-sufficiency, and CO_2 reduction can be included. The proposed strategy considers the following metrics for evaluation: cost savings, grid dependence index, self-consumption, and self-sustainability, fairness index for VS, and CO_2 reduction.

Cost Savings (*CS*) of the community represent the ratio between the difference between the final payment of the scenario and the initial payment:

$$CS_{EC} = \frac{Pay_{EC} - Pay_{EC,0}}{Pay_{EC,0}} \times 100. \tag{22}$$

Grid Dependence Index (*GDI*) measures the extent to which the community relies on the electrical grid for its energy needs. It is typically calculated as the ratio of the electricity imported from the grid (or the uncovered demand from local generation) to the total electricity consumed. A higher GDI indicates greater dependence on the grid, while a lower GDI suggests higher self-sufficiency:

$$GDI_{EC} = \frac{\sum_t D_{EC}^t}{\sum_t C_{EC}^t} \times 100. \tag{23}$$

Self-Consumption Index (*SCI*) quantifies the proportion of locally generated renewable energy that is consumed directly by the community rather than being exported to the grid. It is expressed as the ratio of self-consumed renewable energy to the total renewable energy generated. A higher SCI indicates more efficient use of locally produced energy:

$$SCI_{EC} = \frac{\sum_t Gu_{EC}^t}{\sum_t G_{EC}^t} \times 100, \tag{24}$$

where Gu_{EC}^t represents the generation used by the community and not injected into the grid and it is calculated as follows:

$$Gu_{EC}^t = \min_t \left(C_{EC}^t, G_{EC}^t \right). \tag{25}$$

Self-Sustainability Index (*SSI*) indicates the ability of the community to meet its energy needs independently of the grid. It is calculated as the ratio of self-produced renewable energy to the total energy consumption over a given period. A higher SSI signifies greater energy independence and resilience:

$$SCI_{EC} = \frac{\sum_t Gu_{EC}^t}{\sum_t C_{EC}^t} \times 100. \tag{26}$$

Fairness Index (*FI*) assesses the equity of the distribution of benefits (such as cost savings and revenue) within the community. It evaluates how fairly the value generated by shared renewable energy resources is allocated among the community members. An FI close to 100 indicates a more equitable distribution, while values significantly different from 1 suggest imbalances in the value share allocation:

$$FI_{EC} = \frac{1}{m} \times \frac{\left(\sum_{i=1}^{m} \frac{VS_i}{\sum_t G_i^t}\right)^2}{\sum_{i=1}^{m} \left(\frac{VS_i}{\sum_t G_i^t}\right)^2} \times 100. \quad (27)$$

In case the allocation method is fair for all scenarios with FI close to 1, the VS index can be used for evaluation. The VS index represents the ratio between the VS and the payment of the members.

CO_2 reduction (ΔCO_2) calculates the amount of CO_2 emissions avoided in each scenario: using PV for covering the electricity consumption (A and B), replacing conventional heating with electric heating (C and E), and replacing conventional cars with EVs (D and E).

For each case, the following values of the Emission Factor (EF, kg CO_2/kWh) are considered: EF for grid electricity (EF_{el}) that can vary depending on the energy mix of each country, which includes the proportions of coal, natural gas, nuclear, hydro, wind, and solar power. In the simulation section, a typical value for the CO_2 emission factor for electricity in Romania is around 0.250 kg CO_2/kWh. For conventional heating using natural gas, EF_H in Romania is 0.181 kg CO_2/kWh, and for conventional transportation, emission factor for gasoline (EF_T) is 2.31 kg CO_2/L. For transportation, a conventional car fuel consumption (ε) of 8 L/100 km can be considered. Thus, the initial CO_2 emissions are calculated as the sum of the initial electricity, gas and gasoline consumption:

$$CO_{2EC,0} = \sum_t C_{EC}^t \times EF_{el} + \sum_t C_H^t \times EF_H + \varepsilon \times \frac{\sum_t km^t}{100} \times EF_T. \quad (28)$$

The total reduction of CO_2 emissions is calculated using the following equation, where $\Delta CO_2(H)$ represents the CO_2 reduction for replacing the conventional heating sources and $\Delta CO_2(EV)$ represents the CO_2 reduction for replacing conventional transportation. Both equations for $\Delta CO_2(H)$ and $\Delta CO_2(EV)$ consider the case in which the generation of the PV systems covers partially the consumption for heating and transportation and subtract the emission for covering the deficit from grid:

$$\Delta CO_2 = \sum_t \left(G_{EC}^t - G_{EV}^t - G_H^t\right) \times EF_{el} + \Delta CO_2(H) + \Delta CO_2(EV), \quad (29)$$

$$\Delta CO_2(H) = \sum_t G_H^t \times EF_H - \sum_t \left(C_H^t - G_H^t\right) \times EF_{el}, \quad (30)$$

$$\Delta CO_2(EV) = \varepsilon \times \frac{\sum_t km^t}{100} \times EF_T - \sum_t \left(C_{EV}^t - G_{EV}^t\right) \times EF_{el}. \quad (31)$$

The CO_2 reduction index (RCO_2) can be expressed by dividing the ΔCO_2 with the initial CO_2 emissions:

$$RCO_{2_{EC}} = \frac{\Delta CO_2}{CO_{2EC,0}} \times 100. \quad (32)$$

The objective function of the strategy (f_{EC}) is to maximize the benefits related to a specific scenario by weighting the features with a corresponding coefficient ($\omega_j \in [0, 1]$). Thus, the objective function is expressed using the following equation:

$$f_{EC} = \omega_1 FI_{EC} + \omega_2 CS_{EC} + \omega_3 SSI_{EC} + \omega_4 SCI_{EC} - \omega_5 GDI_{EC} + \omega_6 \frac{1}{PBP} \times 100 + \omega_7 RCO_{2_{EC}}. \quad (33)$$

The weights can be adjusted depending on the community preferences. In case monetary aspects prevails, then ω_1 and ω_6 are set to 1, while the rest are lower. In case the sustainability prevails, then ω_2, ω_3, and ω_4 will have higher values.

The data used to support the simulation of each scenario, including the source of load profiles, heating and transport assumptions, and weather data, is further detailed in Section 4.

4 Results

4.1 Input Data and Assumptions

For the simulations, we use the *Smart Data Set for Sustainability*,[6] which contains real electricity consumption data collected from 114 residential apartments over a 3-year period. The dataset has a 15-min resolution and includes timestamped individual consumption values. This high temporal granularity allows for detailed modelling of demand patterns and compatibility with ToU tariff structures. Since the dataset only contains individual electricity consumption, we assume shared (communal) consumption, such as for elevators, hallway lighting, or common utilities, to be approximately 3% of the total, based on typical values reported in residential building studies. Additionally, heating and Electric Vehicle (EV) consumption profiles are

6 https://traces.cs.umass.edu/index.php/Smart/Smart

Table 2 Parameters used in the simulations

Parameter	Value/assumption	Unit	Source/notes
Installation cost of PV systems	35,000 lei per 10 kWp	lei (€ 7000)	Market estimate for medium-scale residential PV systems in Romania
Installation cost of centralized heat pump (HP)	1,710,000 lei (≈€ 342,000)	lei	Based on € 3000 per household for centralized installation
Installation cost of level 3 EV charging station	100,000 lei	lei (€ 20,000)	Community-scale DC fast charger, including installation and permitting
ToU electricity price	1 lei	lei/kWh (€ 0.20/kWh)	Romanian national average, 2023 (source: ANRE[9])
FiT electricity price	0.3 lei	lei/kWh (€ 0.06/kWh)	Romanian national average, 2023 (source: ANRE)
Gas price	0.31 lei	lei/kWh (€ 0.06/kWh)	Romanian national average, 2023(source: ANRE)
Petrol price	7 lei	lei/L (€ 1.4/L)	Romanian national average, 2023 (source: ANRE)
Emission factor—electricity (EF_{el})	0.250	kg CO_2/kWh	Romanian national grid mix (source: Romanian TSO[10] and IPCC[11])
Emission factor—heating (EF_H)	0.181	kg CO_2/kWh	For natural gas; national average value
Emission factor—transport (EF_T)	2.31	kg CO_2/L	IPCC & EU vehicle emissions data
Conventional car fuel consumption (ε)	8	liters/100 km	Romanian fleet average (source: Eurostat[12], 2023)
EV consumption (EC_{EV})	15	kWh/100 km	Based on compact/mid-size EV models
EV charging efficiency (η_{EV})	0.9	-	charging efficiency ~ 90%
Annual distance per member (km^{365})	10,000	km	Survey-based approximation
Charging energy per vehicle per year	~ 1650	kWh/year	Based on 10,000 km/year × 15 kWh/100 km ÷ 0.9 efficiency
Insulation coefficient (k)	0.2	W/m^3 °C	EN ISO 52016–1:2017 standard for building thermal performance
Avg. dwelling volume (V_i)	175	m^3	Assumes 70 m^2 × 2.5 m ceiling height
Indoor temperature	24	°C	Estimated comfort level
Heat pump type	Air-to-water	-	Most commonly installed HP in Romanian apartment blocks
COP	3.2	-	Assumed constant average value (EHPA 2023)
Specific PV yield	1200	kWh/kWp/year	Based on Global Solar Atlas estimates for Bucharest region
Average gas consumption per member (m^3)	Winter, 65; Spring and Autumn, 30	m^3	Romanian national average, 2023(source: ANRE)
Weights of the objective function	All set to 1	-	-

[9]https://arhiva.anre.ro/ro/info-consumatori/comparator-de-tarife

[10]https://www.transelectrica.ro/

[11]https://www.ipcc-nggip.iges.or.jp/public/gp/bgp/2_3_Road_Transport.pdf

[12]https://ec.europa.eu/eurostat

synthetically generated based on actual weather data (temperature time series) and standardized assumptions regarding building insulation, household volume, heating demand per degree-day, and EV driving patterns. These assumptions and parameter values are detailed in Table 2.

Agent-Based Modelling (AGM) using Mesa[7] Python library is used to simulate the behavior of the members, having an approach as close as possible to reality. Also, the temperature values from the weather dataset are used to calculate the heating requirement using Eq. (19), considering an inside temperature of 24°C. The values of the parameters used in the simulations are centralized in Table 2.

For each scenario, the synthetic data is obtained using the agents to extract individual consumption from the initial dataset, calculate the heating requirement, generate the schedule of the HP, generate the daily distance to

[7] https://mesa.readthedocs.io/en/stable/

Table 3 Initial payment of the community

Initial payments	Year	Annual average per member	Winter month	Spring and Autumn month	Summer month	Monthly average per member
Initial payment (lei)	**1,221,984.19**	**10,719.16**	**117,504.74**	**100,616.24**	**95,967.00**	**893.26**
Payment for electricity (lei)	**473,877.75**	4156.82	42,254.25	36,430.00	42,845.00	346.40
Payment for gas heating (lei)	**110,642.44**	970.55	22,128.49	11,064.24		80.88
Payment for transportation using gasoline (lei)	**637,464.00**	5591.79	53,122.00	53,122.00	53,122.00	465.98

Table 4 Initial CO_2 emissions for electricity, gas and transportation

Emissions type	Initial CO_2 emissions (kg)
Electricity	118,469.4
Gas	64,600.9
Petrol	210,361.5
Total	393,431.8

be covered by the EV charging, calculate the surplus and deficit, and trade it on LEM in case of scenario F. The individual values are aggregated at the community level, and the metrics from Section 3.2 are calculated.

The initial payment before installing PV systems ($Pay_{EC,0}$) is detailed in Table 3 and includes the electricity payment using ToU, cost for heating using gas and payment for transportation using gasoline. The total payments are obtained based on the parameters from Table 2.

The emissions factors for electricity, gas, and petrol are applied to calculate the initial CO_2 emissions (Table 4).

4.2 Scenario A

In this scenario the rated power of the PV systems is 10 kWp determined based on the shared electricity consumption that needs to be covered, calculated with Eq. (5) and the investment cost ($Cost_{EC}$) is 35,000 lei (€7000).

The agents extract the records of the 114 apartments and determine the shared consumption as 3% from the individual consumption. The values are aggregated at the community level, and the surplus and deficit are obtained using Eqs. (3–4). The total payment of the community is determined based on Eq. (2), and the metrics are centralized in Table 5.

In this case, the cost savings is 1%, without value share since only the shared consumption is considered and the self-sustainability metrics indicate a very high dependence on the main grid. CO_2 reduction is insignificant. The payback period of the initial investment is less than 3 years.

Two days are selected for representation of Scenario A, one day in winter in which the generation partially covers the shared consumption and one day in summer when the generation exceeds the shared consumption (Fig. 2).

4.3 Scenario B

The installed power increased to 100 kWp and the initial cost for installation is 350,000.00 lei (€70,000). In this case, the agents extract the individual consumption from the initial dataset, calculate the shared consumption as in Scenario A, and the values are aggregated at the community level. The total payments still include the initial costs for heating and transportation. The metrics are centralized in Table 6.

The electricity cost reduction is significant, the cost savings for the entire year are more than 10%, and the payback period is less than 3 years, although the initial cost is 10 times higher than Scenario A.

Figure 3 depicts the same days from winter and summer in case of Scenario B.

4.4 Scenario C

In this scenario, the heating is shifted from the conventional sources (gas) to electric heating by installing a high-capacity heat pump for centralized heating in the building. The investment cost with the installation of the HP is very high (€342,000) since these technologies are still very expensive. The cost per apartment is estimated to be €3000, including labor, pipes, and fittings. Thus, the scenario assumes that all apartments are shifting from gas to HP, and the initial investment covers all their expenses. So, the final payment for gas is 0. The rated power of the PV system is increased to 200 kWp, and the investment cost is €140,000. The final payments in this scenario include the initial transportation cost for petrol. The metrics are centralized in Table 7.

The savings are almost double compared to Scenario B and the self-sufficiency metrics are considerably improved. The CO_2 reduction is also significant, more than 25% compared to the initial state. The objective function is more than 100 points, revealing that Scenario C is a better strategy compared to A or B. The drawback of this scenario is related to the high payback period (more than 9 years) due to the fact that HP is still expensive.

Table 5 Metrics for Scenario A

Metrics	Year	Annual average per member	Winter month	Spring/autumn month	Summer month	Monthly average per member
Electricity payment of the members (lei)	460,257.75	4037.3	41,074.25	35,350.0	41,645.00	336.45
Electricity payment of the EC (lei)	460,257.75	4037.3	41,074.25	35,350.0	41,645.00	336.45
Savings (lei)	13,620.00	119.47	1180.00	1080.0	1200.00	9.96
Value share (lei)	0	0	0	0	0	0
VS index (%)	0	0	0	0	0	0
Cost savings (CS) (%)	1.1	1.1	1.0	1.1	1.3	1.1
Self-Sustainability Index (SSI) (%)	4	4	4	4	4	4
Self-Consumption Index (SCI) (%)	97	97	97	97	97	97
Grid dependence index (%)	98	98	99	98	95	98
Payback period (years)	2.57					
Final CO_2 emissions (kg)	90,026.88					
CO_2 reduction (kg)	3405.00					
CO_2 reduction index	0.87					
Objective function	43.88					

The same days from winter and summer in case of Scenario C are depicted in Fig. 4. As noticed, the electricity required for heating in winter is almost 25% of the total electricity consumption, while in summer the electricity used for cooling is less than 10% of the total load.

Figure 4 indicates that the generation exceeds the load, especially during the summer days; therefore, the same installed power can be used in Scenarios D, E, and F.

4.5 Scenario D

In this case, the same investment cost for the PV system is used, and the EV charging station is added. The cost is estimated to be 100,000 lei (€20,000) for a Level 3 charging station with 80 kW. The scenario assumes that the members cover their own car exchange costs (replacing fossil fuel cars with hybrid or EVs). Thus, the final payment for the transportation using petrol is 0. The EVs are charged from the building station, and the electricity consumed for charging is allocated to the members using individual cards or PINs. Therefore, the consumption is supported by the member who charges the car. The final payment includes the gas cost since the conventional sources are used for the heating in this scenario. The metrics are centralized in Table 8.

The scenario provides the best values compared to Scenarios A, B, and C for the payback period (approx. 1 year),

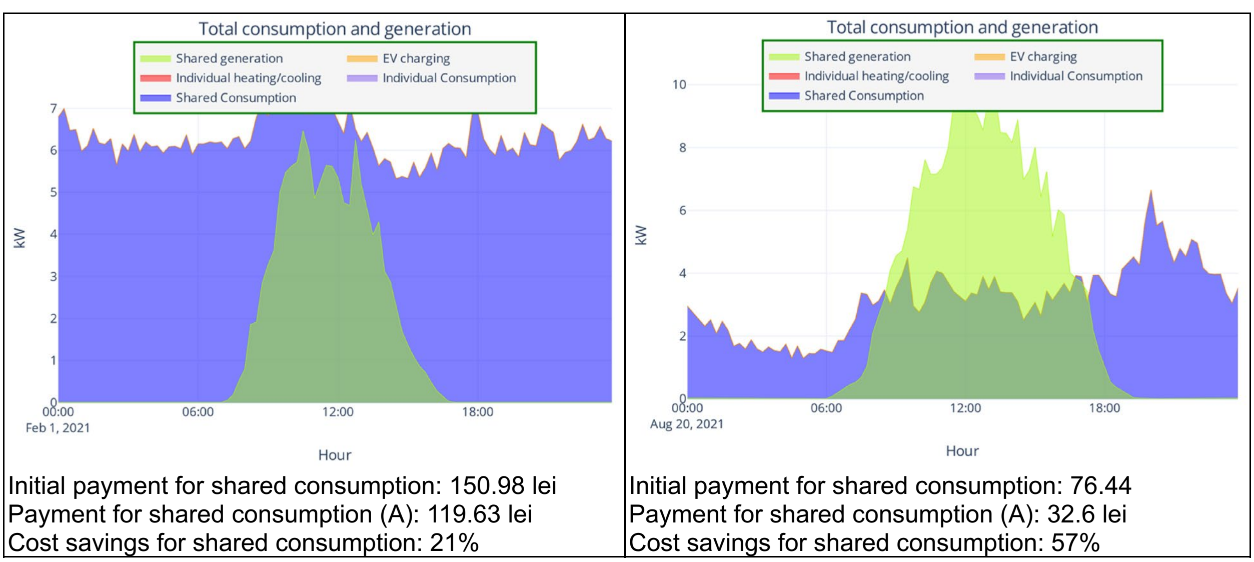

Fig. 2 Shared consumption and generation in Scenario A

Table 6 Metrics for Scenario B

Metrics	Year	Annual average per member	Winter month	Spring/autumn month	Summer month	Monthly average per member
Electricity payment of the members (lei)	351,220.92	3080.9	35,860.00	24,769.00	31,675.64	256.74
Electricity payment of the EC (lei)	326,410.77	2863.3	35,348.00	23,186.00	27,083.59	238.60
Savings (lei)	147,466.98	1,293.57	6906.25	13,244.00	15,761.41	107.80
Value share (lei)	24,802.77	217.6	511.53	1582.00	4592.06	18.13
VS index (%)	7.06	7.06	1.43	6.39	14.50	7.06
Cost savings (CS) (%)	12.07	12.07	5.88	13.16	16.42	5.88
SSI (%)	22.00	22.0	14	27	20	22.0
SCI (%)	64.00	64.0	85	60	51	64.0
Grid dependence index (%)	70.75	70.8	83.91	67.19	64.71	70.8
Payback period (years)	2.37					
Final CO_2 emissions (kg)	356,565.14					
CO_2 reduction (kg)	36,866.75					
CO_2 reduction index	9.37					
Objective function	104.94					

cost savings, self-sufficiency, and CO_2 reduction. The savings are considerably high, almost 60% of the initial payment, and the value share that is distributed to the members after net metering the EC is double compared to Scenario C.

In Fig. 5 the shapes of the load and generation for Scenario D are depicted for the same days in winter and summer. The generation on peak is enough to cover the charging of the EVs and still there is availability for extra load (e.g. heating in the next scenarios).

As noticed from Fig. 5, the EVs are not always charged during the generation peak, since the car owners are not at home at noon, but other load can be shifted and optimized to increase the consumption during the PV availability and decrease it in the afternoon or evening, as demonstrated in [5].

4.6 Scenarios E and F

In these scenarios the heating and transportation are shifted from conventional sources to electricity; therefore, the final payment includes only the electricity payment. To compare Scenarios E and F with Scenarios C and D, the same investment cost for PV systems, HP and EV charging are maintained, although a higher rated power may be required (up to 250 kWp). To enable LEM trading, the agents are using

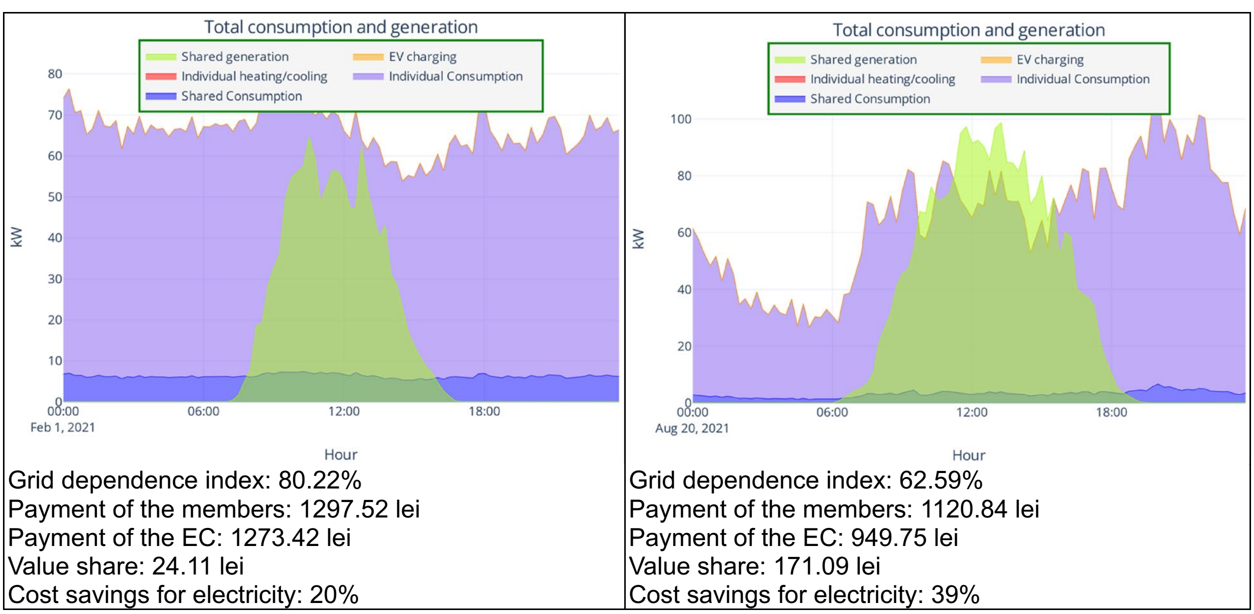

Fig. 3 Electricity consumption and generation in Scenario B

Table 7 Metrics for scenario C

Metrics	Year	Annual average per member	Winter month	Spring/autumn month	Summer month	Monthly average per member
Electricity payment of the members (lei)	355,524.00	3118.6	47,184.00	21,989.00	27,346.00	259.89
Electricity payment of the EC (lei)	324,528.00	2846.7	45,626.00	20,022.00	22,506.00	237.23
Savings (lei)	259,992.19	2,280.63	18,756.74	27,472.24	20,339.00	190.05
Value share (lei)	30,996.00	271.9	1,558.00	1967.00	4840.00	22.66
VS index (%)	8.72	8.72	3.30	8.95	17.70	8.72
Cost savings (CS) (%)	21.28	21.28	15.96	27.30	21.19	21.28
SSI (%)	34.00	34.0	17	34	35	34.0
SCI (%)	63.00	63.0	68	63	40	63.0
Grid dependence index (%)	59.90	59.9	79.54	67.19	56.29	59.9
Payback period (years)	9.27					
Final CO_2 emissions (kg)	291,493.54					
CO_2 reduction (kg)	101,938.35					
CO_2 reduction index	25.91					
Objective function	103.79					

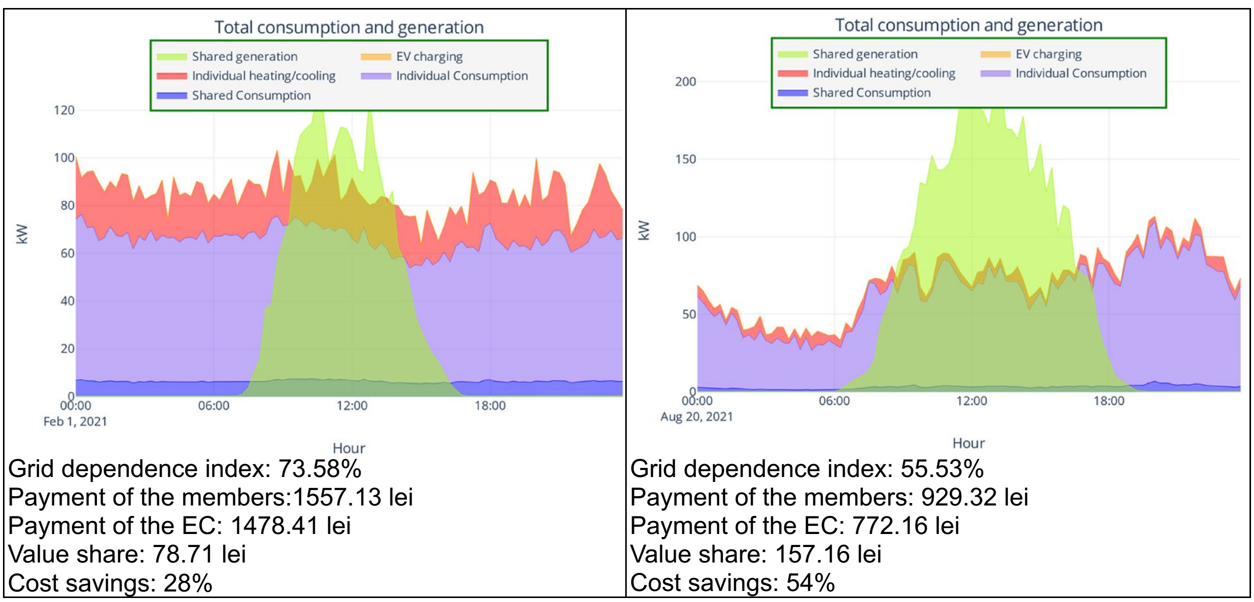

Fig. 4 Electricity consumption and generation in Scenario C

the PyMarket[8] Python library, and the Uniform Price mechanism is used for market clearing. The metrics are calculated similarly for both scenarios; only the final payment of the members and the value share differ in Scenario F compared to Scenario E. Let us assume that the same fairness index (FI) is used in both scenarios instead of the VS index; thus, the same values for the objective functions are obtained for both scenarios. The metrics are centralized in Table 9.

Both scenarios have very good self-sufficiency metrics, CO_2 reduction, and cost savings values, but the payback period is higher than Scenario D due to the high investment cost for installing the HP. Therefore, the objective function has a lower value than in Scenario D.

As noticed from Fig. 6, the generated power at peak covers the load, so the PV system is sized correspondingly, and no additional investment is needed.

In short, our research modeled six operational scenarios for ECs, each incorporating increasing levels of energy integration, ranging from shared electricity

8 https://pymarket.readthedocs.io/en/latest/

Table 8 Metrics for scenario D

Metrics	Year	Annual average per member	Winter month	Spring/ Autumn month	Summer month	Monthly average per member
Electricity payment of the members (lei)	454,376.40	3985.8	46,335.80	33,097.00	38,929.00	332.15
Electricity payment of the EC (lei)	381,421.20	3345.8	39,733.40	28,381.00	30,645.00	278.82
Savings (lei)	729,920.55	6,402.81	55,642.85	61,171.00	65,322.00	533.57
Value share (lei)	72,955.20	640.0	6602.40	4716.00	8284.00	53.33
VS index (%)	16.06	16.06	14.25	14.25	21.28	16.06
Cost savings (CS) (%)	59.73	59.73	47.35	60.80	68.07	59.73
SSI (%)	24.00	24.0	22	26	22	24.0
SCI (%)	47.00	47.0	66	42	38	47.0
Grid dependence index (%)	62.61	62.6	73.2	61.37	54.49	62.6
Payback period (years)	1.10					
Final CO_2 emissions (kg)	159,956.21					
CO_2 reduction (kg)	233,475.67					
CO_2 reduction index	59.34					
Objective function	234.76					

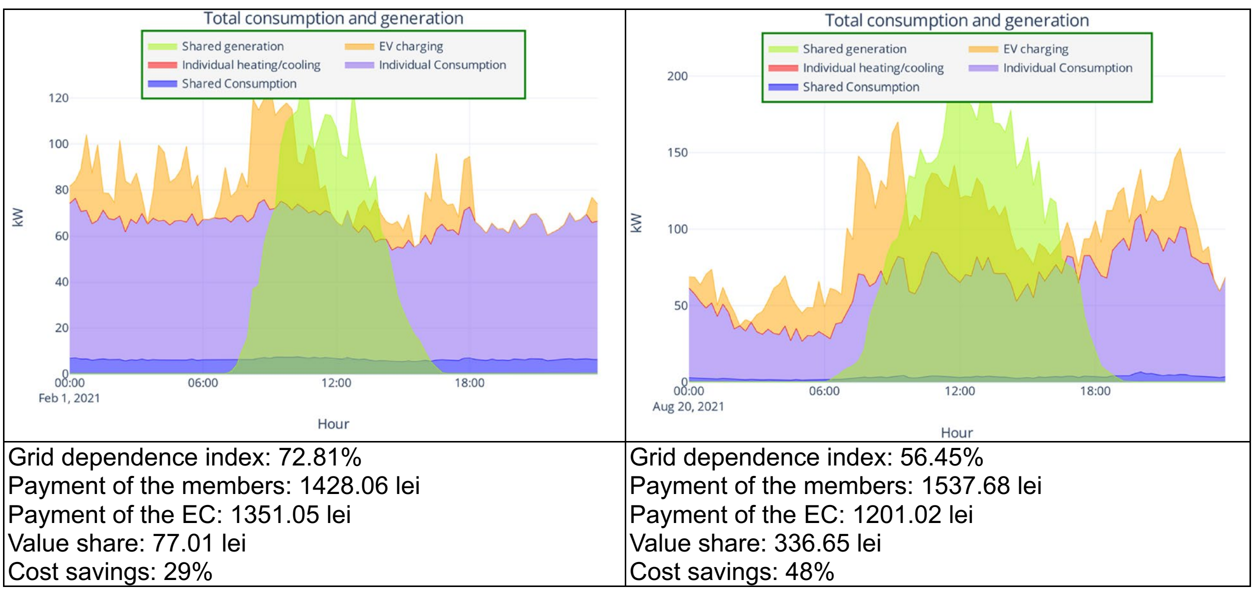

Fig. 5 Electricity consumption and generation in Scenario D

consumption (Scenario A) to full electrification of heating and transportation combined with participation in Local Energy Markets (Scenario F). Simulations were performed using agent-based modeling on a high-resolution, real-world dataset (15-min intervals, 3 years, 114 apartments). The findings reveal several scientific insights and implications:

- *Scenario A: baseline integration—shared consumption only*

  ○ *Self-sufficiency and savings are minimal.* With only 10 kWp PV systems covering communal electricity (~3% of total), cost savings are marginal (~ 1%) and CO_2 reductions are negligible (< 1%).

  ○ *Scientific implication.* Low-impact baseline confirms that shallow EC integration yields limited environmental and economic benefits, but offers a short payback period (~ 2.6 years), which may serve as a low-risk entry point for EC adoption.

- *Scenario B: full electricity coverage for households*

  ○ *Substantial increase in cost-effectiveness.* With a 100 kWp PV installation, cost savings increase to ~ 12%,

Table 9 Metrics for scenarios E and F

Metrics	Year	Annual average per member	Winter month	Spring/Autumn month	Summer month	Month avg per member
Electricity payment of the members (lei)—*Scenario E*	564,180.50	4949.0	58,608.29	35,577.38	39,719.79	412.41
Electricity payment of the members after LEM (lei)—*Scenario F*	488,176.55	4282.3	56,515.96	31,102.81	32,428.25	356.85
Electricity payment of the EC (lei)	478,259.50	4195.3	56,496.85	31,066.12	32,333.7	349.60
Savings	743,724.69	6523.90	61,007.89	69,550.12	63,633.30	543.66
Value share without LEM (lei)—*Scenario E*	85,917.35	753.7	2111.44	4511.26	7386.08	62.81
Value share with LEM (lei)—*Scenario F*	9917.05	87.0	19.11	36.7	94.54	7.25
VS index without LEM (%)—*Scenario E*	15.23	15.23	3.60	12.68	18.60	15.23
VS index with LEM (%)—*Scenario F*	2.03	2.03	0.03	0.12	0.29	2.03
Cost savings (%)	60.86	60.86	51.92	69.12	66.31	60.86
Self-sustainability index (SSI) (%)	19.00	19.0	14	25	23	19.0
Self-consumption index (SCI) (%)	46.00	46.0	69	43	41	46.0
Grid dependence index (%)	66.54	66.5	81.68	63.08	59.14	66.5
Payback period (years)	3.37					
Final CO_2 emissions (kg)	119,564.88					
CO_2 reduction (kg)	273,867.01					
CO_2 reduction index	69.61					
Objective function	173.79					

and CO_2 reductions to ~ 9.4%, while maintaining a short payback period (~ 2.4 years).
○ *Scientific insight.* Expanding EC coverage from communal to individual household loads offers significantly greater returns due to better alignment of generation and consumption, illustrating non-linear benefits of scale in distributed energy systems.

- *Scenario C: electrification of heating*

  ○ *High environmental gains*, *but long payback.* Shifting from gas to centralized heat pumps leads to a 26% reduction in emissions, but the investment cost triples, pushing the payback to ~ 9.3 years.
  ○ *Trade-off identified.* Electrification of heating is environmentally effective but economically constrained under current market conditions, underscoring the need for policy incentives or falling technology costs for broader adoption.

- *Scenario D: integration of electric mobility*

  ○ *Best economic and environmental performance.* Replacing petrol cars with EVs charged from PV yields nearly 60% cost savings, the shortest payback (~ 1.1 years), and greater than 59% CO_2 reduction.
  ○ *Key scientific finding.* Electric mobility, when powered locally via RES, provides the highest value proposition for ECs, reinforcing the synergistic potential of sector coupling (transport-electrification-energy).

- *Scenario E: full energy electrification without market interaction*

  ○ *Broad integration with good but plateauing benefits.* Electrifying both heating and transportation boosts emissions savings (~ 69.6%) and cost savings (~ 60.9%), but the payback period (~ 3.4 years) is longer than Scenario D.
  ○ *Scientific implication.* Integration of all energy services increases complexity and CAPEX, leading to diminishing marginal returns in cost savings, highlighting the need for intelligent optimization strategies.

- *Scenario F: participation in local energy markets* (*LEM*)

  ○ *Mixed impact from market integration.* While LEM trading reduces the individual payment, it also significantly lowers the internal value share (from ~ 15.2 to ~ 2%), suggesting benefits from market participation are not evenly distributed.
  ○ *Scientific insight.* Market mechanisms improve collective efficiency but may reduce fairness or perceived gains for individual members, suggesting that governance and redistribution mechanisms must be carefully designed in ECs engaging in LEM.

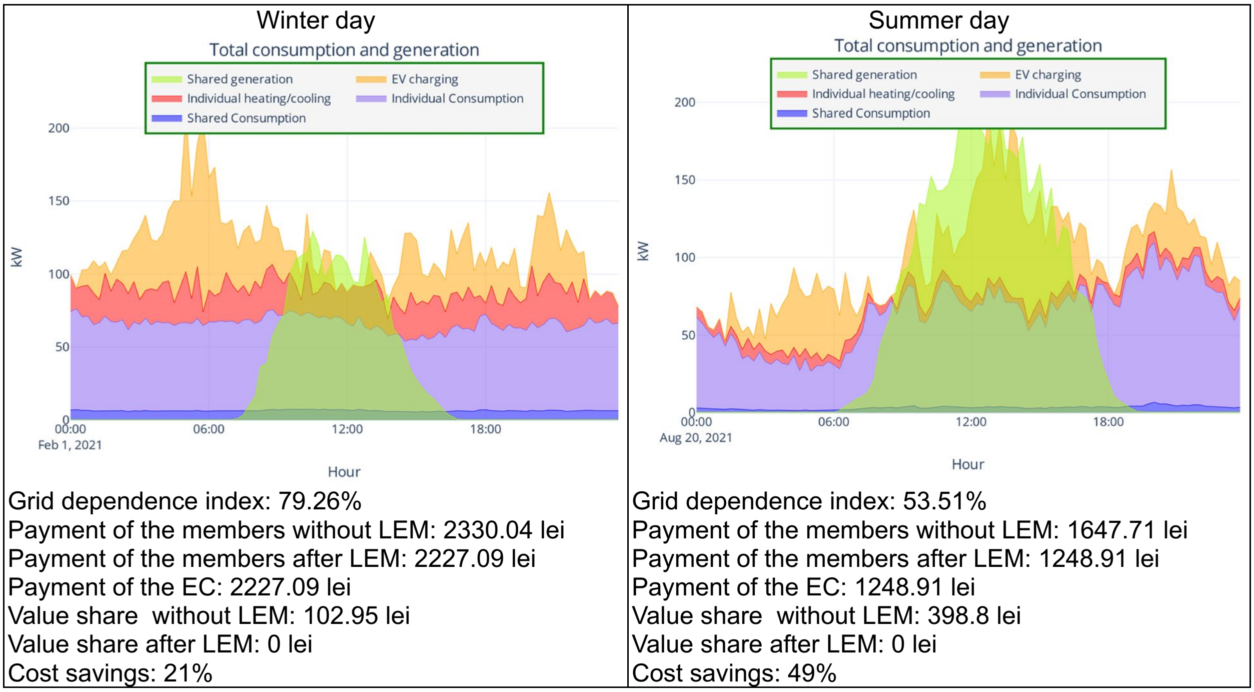

Fig. 6 Electricity consumption and generation in Scenarios E and F

The main cross-cutting implications are summarized. *For policymakers*, the findings underscore that a one-size-fits-all approach to EC policy is inadequate for CEE contexts. Incentive structures should be stratified according to the complexity of EC scenarios, while technical assistance programs need to be adapted to specific archetypes, distinguishing between urban and rural communities, as well as between new residential developments and retrofitted legacy buildings. *For EC initiators*, electrification of transport emerges as the most promising starting point due to its rapid payback and substantial cost savings, whereas electrifying heating systems requires greater upfront investment and deeper community engagement. In this regard, initiating simpler EC projects can serve as a trust-building mechanism in communities where civic participation is historically low. *For researchers*, this research demonstrates the value of using high-resolution data and agent-based modeling to assess EC feasibility. Future research should further integrate socio-technical dynamics, including user adoption behavior, collective decision-making processes, and more psychological or cultural barriers, to better inform the design and scalability of ECs in transitional energy systems.

5 Conclusions

This paper examines diverse urban and rural archetypes and various operational scenarios for energy communities. It proposes a methodology for developing a strategy to select the optimal scenario based on factors such as self-sufficiency, payback period, cost savings, value share allocation, and CO_2 reduction. The proposed methodology is used to assess the investment impact, heating, and transportation transition using an incremental approach from Scenario A to Scenario F. The simulations involve 114 apartments over a 1-year period, focusing on optimizing electricity consumption, heating, and transportation. Various scenarios (A to F) incorporate PV systems from small size (10 kWp) used to cover only the shared consumption to medium size (200 kWp) for covering electricity consumption including heat pumps and EV charging stations to assess their impact on cost savings, self-sufficiency, and CO_2 emissions.

Scenario A represents the starting point for ECs and demonstrates the viability of the PV systems. Although the rated power is small, the generation exceeds the shared consumption, and members may become aware of the potential and benefits of the local generation. In Scenarios C, D, and E, larger PV systems significantly reduce costs and CO_2 emissions but require higher initial investments. Scenarios with higher rated PV power (100 kWp and 200 kWp) show better savings and self-sufficiency metrics. Switching to electric heating (Scenario C) offers substantial CO_2 reductions but comes with high upfront costs and longer payback periods. Introducing EV charging (Scenario D) drastically improves savings and reduces emissions, showcasing the potential benefits of electrifying transportation.

From the point of view of cost–benefit analysis, Scenario D is the most cost-effective with the shortest payback period and high CO_2 reduction. In case Scenario D includes the cost of replacing the conventional cars with EVs, the payback period is longer, but it is the choice of the members to replace their cars, and the scenario evaluates only the viability of the integration between PV systems and EVs charging. Scenarios E and F demonstrate the highest CO_2 reductions, but require longer to recoup the initial investments, considering that the HP installation cost is included. A phased approach can be considered, starting with Scenario B (100 kWp) for moderate savings and emissions reduction and gradually introducing EV charging (Scenario D) for higher savings and shorter payback periods and then shifting towards electric heating using HP (Scenario E) for their long-term benefits. Finally, leverage LEM trading to maximize savings among EC members, as seen in Scenario F. In all scenarios, to improve self-sufficiency, several incentives for shifting the load to hours with available generation should be used. Thus, the generation can cover more consumption, and the final payment decreases considerably.

The methodology and the insights of the simulations highlight the importance of strategic planning and phased investments to achieve optimal sustainability outcomes in EC management. The objective function can be customized using the weights, depending on the community interests, to provide the best scenario that fits the needs of the members.

Author Contribution Adela Bâra: Methodology, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing-Original Draft, Writing-Review and Editing, Visualization, Supervision. Simona-Vasilica Oprea: Conceptualization, Formal analysis, Investigation, Writing-Original Draft, Writing-Review and Editing, Visualization, Project administration. Corina Murafa: Formal analysis, Investigation, Writing-Original Draft, Writing-Review and Editing, Validation.

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Declarations

Ethics Approval Not applicable.

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